

JOUTSA LEIVONMAKI SAVENAHÖ, Finland

WMO: 027710

Lat: 61.879N

Lon: 26.095E

Elev: 146

StdP: 99.58

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.8	(c) -21.7	(d) -27.4	(e) 0.3	(f) -24.7	(g) -24.0	(h) 0.4	(i) -21.7	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	27.0	18.5	24.8	17.7	22.9	16.5	20.3	24.6	19.0	23.0	17.8	21.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.8	21.9	17.6	12.8	20.8	16.4	11.9	19.3	59.0	24.9	54.8	23.1	50.8	21.5	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -26.1	(f) 28.9	(g) 4.3	(h) 2.5	(i) -29.2	(j) 30.7	(k) -31.7	(l) 32.2	(m) -34.1	(n) 33.6	(o) -37.1	(p) 35.4		
(4) N/A	(5) N/A	(6) N/A	(e) -26.3	(f) 21.2	(g) 4.2	(h) 2.0	(i) -29.3	(j) 22.6	(k) -31.8	(l) 23.8	(m) -34.2	(n) 24.9	(o) -37.3	(p) 26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-8.7	-6.4	-2.9	3.1	10.0	13.7	16.8	14.9	10.2	3.7	0.0	-4.3	(6)
(7)		DBStd	9.62	6.93	6.40	4.58	3.61	4.49	3.77	3.55	3.03	3.32	3.56	4.23	5.95	(7)
(8)		HDD10.0	2692	579	459	400	208	56	9	1	2	38	196	301	443	(8)
(9)		HDD18.3	5186	838	693	658	457	259	147	74	113	243	453	551	701	(9)
(10)		CDD10.0	589	0	0	0	2	57	119	211	155	45	1	0	0	(10)
(11)		CDD18.3	41	0	0	0	0	2	6	26	7	0	0	0	0	(11)
(12)		CDH23.3	357	0	0	0	0	32	74	204	47	0	0	0	0	(12)
(13)		CDH26.7	66	0	0	0	0	3	12	42	9	0	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	636	50	37	34	34	43	63	81	70	53	62	56	52	(15)
(16)		PrecMax	813	91	83	78	63	114	98	185	137	136	129	118	97	(16)
(17)		PrecMin	514	9	2	3	10	9	25	21	19	9	14	5	16	(17)
(18)		PrecStd	74	22	20	15	15	24	21	37	33	29	29	25	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.4	9.2	18.6	26.7	28.3	30.2	27.9	20.3	12.9	9.6	6.7	(19)
(20)		MCWB	2.6	2.7	4.1	10.3	15.5	18.2	20.2	20.5	15.8	11.3	8.7	6.1	6.1	(20)
(21)		2%	DB	1.6	2.5	6.0	14.6	23.6	25.7	27.7	24.2	18.2	11.2	7.6	4.0	(21)
(22)		MCWB	1.2	1.7	2.5	8.1	14.3	17.1	19.6	18.1	14.7	10.1	7.2	3.6	3.6	(22)
(23)		5%	DB	0.6	1.4	4.0	11.7	21.1	23.1	25.7	22.2	16.4	10.3	6.2	2.6	(23)
(24)		MCWB	0.3	1.0	1.7	6.4	13.0	15.6	19.0	16.8	13.8	9.5	5.9	2.2	2.2	(24)
(25)		10%	DB	-0.1	0.6	2.5	9.4	18.5	20.7	23.5	20.2	15.0	9.0	5.0	1.5	(25)
(26)		MCWB	-0.2	0.3	1.0	5.3	11.8	14.7	18.0	15.9	13.5	8.2	4.8	1.2	1.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.8	4.9	11.1	17.1	19.7	22.2	21.4	16.5	11.7	8.9	6.2	(27)
(28)		MDCB	2.8	3.8	6.9	16.9	22.7	25.0	26.7	25.8	19.1	12.2	9.2	6.4	6.4	(28)
(29)		2%	WB	1.2	1.9	3.2	9.0	15.6	18.0	20.8	19.3	15.5	10.6	7.2	3.7	(29)
(30)		MDCB	1.4	2.4	5.1	13.3	21.1	23.3	25.2	22.7	17.2	10.9	7.5	4.0	4.0	(30)
(31)		5%	WB	0.4	1.1	2.3	7.1	14.2	16.8	19.9	18.0	14.5	9.6	6.0	2.2	(31)
(32)		MDCB	0.5	1.4	3.6	10.8	19.6	21.2	24.1	20.5	15.8	10.1	6.2	2.4	2.4	(32)
(33)		10%	WB	-0.2	0.4	1.3	5.6	12.6	15.6	18.7	16.8	13.6	8.2	4.8	1.2	(33)
(34)		MDCB	0.0	0.6	2.2	8.9	17.6	19.7	22.5	19.2	14.7	8.9	5.0	1.4	1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	5.5	7.7	9.2	11.2	10.2	9.6	8.7	7.0	5.1	3.3	3.8	(35)
(36)		MCDBR	3.0	4.4	8.8	13.6	15.1	13.5	12.2	11.4	9.0	6.7	4.5	3.9	3.9	(36)
(37)		MCWBR	3.0	3.8	6.1	7.6	7.1	6.0	5.3	5.8	5.8	5.2	4.3	4.0	4.0	(37)
(38)		5% WB	MCDBR	3.2	4.1	7.1	11.9	13.2	12.0	11.0	9.3	7.8	5.9	4.5	3.8	(38)
(39)	MCWBR	3.2	3.7	5.3	7.2	6.6	5.8	5.5	5.3	5.4	5.1	4.4	3.9	3.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.241	0.288	0.316	0.355	0.353	0.349	0.373	0.358	0.334	0.315	0.262	0.357	(40)	
(41)		taud	2.057	2.148	2.225	2.351	2.442	2.479	2.429	2.459	2.516	2.418	2.152	2.622	(41)	
(42)		Ebn at Noon	443	664	790	834	869	878	843	826	782	650	415	287	(42)	
(43)		Edh at Noon	37	69	91	100	99	97	100	89	70	54	33	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.21	0.79	2.28	3.68	4.92	5.23	5.06	3.90	2.25	0.92	0.24	0.11	(44)	
(45)		RadStd	0.04	0.18	0.28	0.41	0.55	0.36	0.51	0.46	0.27	0.16	0.04	0.01	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	-156	N/A	N/A	N/A

Nomenclature: See separate page